MILITARY SPECIFICATION

GENERATING AND CHARGING PLANT, OXYGEN - NITROGEN, SKID-MOUNTED, A/E26A-8 AND A/E26A-20

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of skid-mounted generating and charging plant for the production of oxygen and nitrogen, designated A/E26A-8, and A/E26A-20.

Type I - Electric motor prime mover using external supply (A/E26A-8).

Type II - Diesel engine prime mover with electric power alternator (A/E26A-20).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solitation, form a part of this specification to extent secified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
P-C-436	Cleaning Compound, Alkali, Boiling Vat (Soak) or Hydrosteam

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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P-C-437	Cleaning Compound, High-Pressure (Steam) Cleaner
BB-N-411	Nitrogen, Technical
CC-M-636	Motor, Alternating Current, (Fractional Horsepower)
CC-M-1807	Motor, Alternating Current, Fractional and Integral Horsepower, (500 HP And Smaller)
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-C-490	Cleaning Methods And Pretreatment Of Ferrous Surfaces For Organic Coatings
TT-E-489	Enamel, Alkyd, Gloss (For Exterior And Interior Surfaces)
TT-P-28	Paint, Aluminum, Heat Resisting (1200°F)
TT-P-636	Primer Coating, Alkyd, Wood And Ferrous Metal
TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tag, Shipping And Stock
VV-F-800	Fuel Oil, Diesel
PPP-B-566	Box, Folding, Paperboard
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed And Lock-Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard Corrugated, Triple-Wall
PPP-B-665	Boxes, Paperboard, Metal Edged And Components
PPP-B-676	Boxes, Set Up
PPP-C-1752	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-P-40	Packaging And Packing Of Hand Tools
PPP-T-60	Tape, Packaging, Water Proof
PPP-T-97	Tape: Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-V-3	Valve, Fittings, And Flanges (Except For Systems Indicated Herein); Packaging Of
MIL-C-104	Crates, Wood; Lumber And Plywood Sheathed, Nailed And Bolted
MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-130	Paper, Wrapping, Laminated And Creped
MIL-B-131	Barrier Material, Water Vaporproof, Flexible, Heat-Sealable
MIL-V-173	Varnish, Mositure And Fungus Resistant (For The Treatment Of Communications, Electronics, And Associated Equipment)
MIL-P-514	Plate, Identification, Instruction And Marking Blank
MIL-T-704	Treatment And Painting Of Material
DOD-D-1000	Drawing, Engineering And Associated List
MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service

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MIL-R-3065	Rubber, Fabricated Parts
MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion Protection On
MIL-R-3593	Refrigeration And Cooling Equipment (Excluding Household Refrigerators), Packaging Of
MIL-C-3600	Compressor, Air And Gas, (Except Oxygen) Packaging Of
MIL-C-3774	Crate, Wood, Open 12,000 And 16,000 Pound Capacity
MIL-S-5002	Surface Treatments And Inorganic Coatings For Metal Surfaces Of Weapon Systems
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-I-6866	Inspection, Penetrant Method Of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-9897	Crate, Slotted Angle, Steel Or Aluminum, For Lightweight Airframe Components And Bulky Items (For Maximum Loads Of 3,000 Pounds)
MIL-E-10062	Engines, Preparation For Shipment And Storage Of
MIL-C-11264	Crates, Wood, Vehicular Assemblies, Reusable Shipping Containers For Tank-Automotive Engines, Transmissions, Differentials, Transfers, Final Drives, And Similar Assemblies
MIL-W-13518	Wood Preservative Tetrachlorophenol And Pentachlorophenol, Surface Sealing Compound
MIL-C-15074	Corrosion Preventive, Fingerprint Remover
DOD-P-16232	Phosphate Coatings, Heavy, Manganese Or Zinc Base (For Ferrous Metals)
MIL-D-16791	Detergent, General Purpose (Liquid, Nonionic)
MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non-Corrosive)
MIL-G-18997	Gages, Pressure, Dial Indicating, Bourdon Tube
MIL-C-23760	Coating, Sprayable, Strippable, Protective, For Preservation And Packaging Of Weapon System And Components; Application Of
MIL-L-26087	Lubricating Oil, Reciprocating Compressor, Ground Support
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, For Steel Surfaces
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid And Gas
MIL-T-27730	Tape, Antiseize, Polytetrafluoroethylene, With Dispenser
MIL-C-52078	Cap-Plug, Cap, And Plug, Protective, Plastic Dust And Moisture Seal
MIL-C-52950	Crate, Wood Open And Covered
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property

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MIL-STD-143	Standards And Specifications Order Of Precedence For The Selection Of
MIL-STD-163	Steel Mill Products Preparation For Shipment And Storage
MIL-STD-649	Aluminum And Magnexium Products Preparation For Shipment And Storage
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing, With Appropriate Test Methods

DRAWINGS

ORDNANCE CORPS

C8987830	Assembly, LOX Coupling, Male Half, All Sizes
B8987832	Gasket, LOX And LN_2, All Sizes
C8987855	Seat, LOX Coupling, 1 Inch
C8987869	LOX Cap, 1 Inch
C8987876	Seat, LN_2 Cap, 1 Inch
C8987888	Assembly, Liquid Nitrogen Coupling, Male Half, All Sizes
C8987890	LN_2 Cap, 1 Inch

*(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 633-78 Zinc On Iron And Steel Electrodeposited Coatings Of

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION

Publication ICl Industrial Control, Standards For

(Copies of the NEMA publication may be obtained upon application to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL BOARD OF FIRE UNDERWRITERS

Pamphlet 70 National Electric Code (Electric Wiring And Apparatus)

(Copies of the NBFU pamphlet may be obtained upon application to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02119.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME Boiler and Pressure Vessel Code

Section VIII Unfired Pressure Vessels

(Copies of the ASME code may be obtained upon application to the American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.)

COMPRESSED GAS ASSOCIATION

Pamphlet V-1 Inlet and Outlet Connections for Compressed Gas Cylinder Valves

(Copies of the CGA pamphlet may be obtained upon application to the Compressed Gas Association, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING.

Safety Code For Mechanical Refrigeration

(Copies of the code may be obtained upon application to the American Society of Heating, Refrigerating and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

DIESEL ENGINE MANUFACTURERS' ASSOCIATION

Standard Practices for Stationary Diesel Engines

(Copies of the above DEMA Standard Practices may be obtained upon application to the Diesel Engine Manufacturers Association, 1 North La Salle St., Chicago, IL 60602.)

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article testing. This specification makes provision for First Article testing.

3.2 Components. The generating and charging plant shall be a complete, self-contained, electric or diesel engine powered unit consisting of all air compression, air refrigeration, heat exchange, air rectification, product servicing, and accessory components needed for compliance with the requirements specified herein. Each component shall include all the parts, controls, instruments, and accessories necessary for its safe and effective operation.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as specified in 3.3.1.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are not suitable

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standard parts. In any case, commercial utility parts such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN without alteration.)

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with varnish conforming to MIL-V-173. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. All metallic surfaces of the generating and charging plant that contact high-purity oxygen or liquid nitrogen, or that are in a location where they will normally be chilled to cause condensation of moisture from the atmosphere, shall be fabricated from corrosion-resistant materials that do not require paint or other organic chemical coatings for protection.

3.4.3 Gasketing and insulating materials. Plastic, rubber-like, rubber, or similar gasketing and insulating materials shall be compounded to insure their suitability for the intended application and, where applicable, their resistance to hydrocarbons or low temperatures.

3.4.3.1 Packing, gaskets, seals, and other nonmetallic components that come into contact with high-purity oxygen shall be compounded of materials compatible with use in a pure oxygen atmosphere, under the conditions to which they will be subjected, and that are not adversely affected or deteriorated by continued use in an oxygen or nitrogen atmosphere.

3.4.4 Rubber goods. Rubber goods such as hoses, seals, insulation or isolation boots, et cetera (etc.), shall be fabricated from rubber compounds conforming to MIL-R-3065.

3.4.5 Combustible materials. Readily combustible materials shall not be used where they can come into contact with high-purity oxygen.

3.5 General design. The generating and charging plant shall be a compact skid-mounted unit that can be readily erected, operated, and maintained in a 20-by 40-foot prefabricated building with floor to eaves height of 14 feet. The building will be erected on a concrete floor with special foundations as specified by the manufacturer.

3.5.1 Generating and charging plant equipment components. Standard, heavy-duty, commercially proven generating and charging plant equipment components shall be used to the maximum practical extent. The generating and charging plant components furnished shall not require special mounting provisions other than tiedown bolts.

3.5.1.1 Each major component will be mounted on skids to facilitate removal of each component without disrupting the remaining equipment installed.

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3.5.2 All electric switchgear, wiring, and lighting equipment (exclusive of building lighting facilities), piping, ducting, controls, instruments, and accessory components needed for installation, operation, and maintenance of the generating and charging plant shall be provided.

3.5.2.1 Electric power supply for Type II generating plant. The Type II generating plant shall not require an external supply of electric power. Provisions shall be made in the plant design to furnish all electrical power needed for lighting the control areas and for operations, servicing, and maintenance of the equipment.

3.5.3 Installation.

3.5.3.1 Components within building. The generating and charging plant design shall provide for installation of the air compressor drive equipment, air compressor, refrigeration compressor and drive, air purification equipment, air processing and rectification equipment, electric control center, instruments, control panels, charging manifold, and other critical components within the specified building area.

3.5.3.2 Components external to building. All other components shall be installed external to the building. Components installed outside the building shall not remove air or heat from or discharge air or heat into the building. Components outside the building include cooling radiators, air-cooled inter-coolers and after-coolers, air-cooled refrigeration condensers, liquid product servicing connections, etc.

3.5.3.3 Installation provisions. Complete plans for layout and foundation specifications for the plant shall be provided.

3.5.4 Shipping sections. When assembled in the specified enclosing building area, the individual shipping sections shall form a complete generating and charging plant capable of the performance requirements specified herein.

3.5.5 Life. The generating and charging plant shall be designed for an operational life of not less than 10,000 hours between component overhaul and to minimize the loss of production time for maintenance, repair, and periodic servicing.

3.5.6 Lubrication. The plant shall be designed to provide for lubrication of all parts subjected to wear. Wherever practicable, such parts shall be designed and constructed to be readily removable. All grease fittings and other lubrication points shall be located to provide maximum accessibility for inspection and replacement. Grease and oil seals shall be designed and located to provide maximum accessibility for inspection and replacement.

3.5.7 Common parts. The generating and charging plant shall be designed to provide for the maximum practicable interchangeability of hardware and fastening devices by using the minimum number of types and sizes of bolts, screws, nuts, washers, and other common hardware.

3.5.8 Detachable fittings. The generating and charging plant shall be designed to include readily detachable and attachable type fittings in the liquid and vapor lines to permit rapid component removal and replacement.

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3.5.9 Protective covers. All cover plates, gaskets, and fittings necessary for protection of contained apparatus during operation, storage, and shipment shall be provided.

3.6 Maintainability. Components and accessories shall be constructed and installed on each individual skid-mounted shipping section to provide for:

a. Ready accessibility to all parts for operation, maintenance, and repairs in the field.

b. The use of conventional, general purpose tools to the maximum extent.

c. Replacement and adjustment of components and accessories with minimum disturbance to other elements of equipment.

3.6.1 Covers or plates that must be removed to accomplish normal operation and maintenance procedures or for frequent component or part removal shall be equipped with substantial quick-disconnect fittings.

3.7 Construction.

3.7.1 Component mounting. Generating and charging plant components shall be securely mounted in a manner that will insure against unnecessary movement and damage during operation and transport. All sub-bases and other fixtures needed for safe and secure component mounting shall be provided.

3.7.1.1 All electrical wiring shall be enclosed in conduit with standard junction-box connectors used at each skid-mounted component. All tubing, lines, and electrical conduit shall be located in protected positions, securely fastened to framework or body structures, and supported to prevent movement and damage during normal plant operation.

3.7.2 Electrical equipment (Type I). Electrical equipment shall be constructed and installed to insure performance in accordance with the requirements specified herein when the generating and charging plant is provided with 440 volts (v) ±10 percent, 3-phase, either 50 or 60 ±3 cycle alternating current (ac). All electrical equipment shall be capable of operating at full load at either frequency without overheating or damage. All switches, circuit breakers, transformers, motor starting devices, and accessories needed for safe and effective operation of the generating and charging plant shall be provided. Electrical components shall be designed, constructed, and installed in accordance with Pamphlet 70 and the National Electrical Manufacturers' Association standards.

3.7.2.1 Operation. All electrical components shall be designed, constructed, and installed to operate from the electrical power supply specified without requiring electrical power generating equipment or batteries as part of the generating and charging plant. Unless otherwise specified by the procuring activity, the electrical equipment shall be connected for operation from a power supply when the generating and charging plant is delivered.

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3.7.2.2 Control voltage. The voltage of electrical power provided at all manually actuated pushbutton stations and switches, and for all control circuits used in operating the generating and charging plant shall not exceed 120V.

3.7.2.3 Electric motors. Electric motors shall conform to CC-M-636 or CC-M-1807 as applicable.

3.7.2.3.1 Motor size. Electric motor size shall be such that the driven load will not exceed the continuous-duty rating of the motor when the equipment is operated under any of the conditions specified herein. The motor shall be constructed to operate continuously at full rated load without overheating or damage.

3.7.3 Electrical equipment (Type II). Electrical equipment shall be constructed and installed to insure performance in accordance with the requirements specified herein. All switches, circuit breakers, transformers, motor starting devices, and accessories needed for safe and effective operation of the generating and charging plant shall be provided. Electrical components shall be designed, constructed, and installed in accordance with Pamphlet 70 and the National Electrical Manufacturers' Association Standards.

3.7.3.1 Control voltage. Same as 3.7.2.2.

3.7.3.2 Electric motors. Same as 3.7.2.3.

3.7.3.2.1 Motor size. Same as 3.7.2.3.1.

3.7.4 Prime mover (Type I). The electric prime mover for the Type I generating and charging plant shall be a single, continuous-duty electric motor designed and constructed to operate on the electric power specified in 3.7.2 and to furnish the motive power required for operation of the air compressor under the conditions specified herein. The motor size shall be such that the total driven load, including losses in the power transmission system, during normal operation of the generating and charging plant under any of the conditions specified herein will not exceed its continuous (24 hours (hrs) per day) output horsepower rating.

3.7.4.1 Motor control. A reduced voltage, magnetically operated control, meeting the requirements of NEMA Publication 1C1 shall be provided for starting and stopping the electric motor prime mover. The motor control shall be equipped with overload protection and undervoltage protection devices of applicable power-handling capacity. The motor control contacts shall pick up and seal at 80 percent of rated voltage or above, and shall not drop out above 60 percent of rated voltage.

3.7.4.2 Control stations. Electric motor prime mover magnetic contactor start and stop button stations shall be mounted in readily visible and accessible locations on the generating and charging plant electric control center, on or near the air processing and separation equipment control panel, and on or near the air compression unit instrument panel. The control station on the air processing and separation equipment control panel or on the air compression unit instrument panel may be omitted if the electric control center is located immediately adjacent thereto and is readily accessible to an operator standing at the instrument or control panel.

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3.7.5 Prime mover (Type II). The Type II prime mover shall be a commercial type diesel engine of a design and construction suited to drive the compressor and auxiliary equipment. The engine size shall be such that the total driven load, including losses in the power take-off, power transmission, and accessory systems will not exceed the continuous (24 hours per day) output horsepower rating recommended by the manufacturer for this type of installation. The brake horsepower rating, governors performance, torsional vibration, material, and workmanship shall be in accordance with DEMA standard practices.

3.7.5.1 Fuel oil. The diesel engine prime mover shall perform as specified herein when supplied with fuel oil conforming to VV-F-800.

3.7.5.2 Equipment. The diesel engine shall be equipped with all manual and automatic controls, instruments, pumps, filters, starting equipment, cooling systems, lubricating systems, power take-offs, and other accessories needed to insure safe and effective operation.

3.7.5.3 Safety devices. Safety devices will be installed on the diesel engine, shut down or audible, to preclude damage to the engine if abnormal operating conditions exceeding those specified by the manufacturer occur (overspeed, low oil pressure, high coolant temperature, overload, electric power failure, etc.)

3.7.6 Controls, instruments, and accessories. All automatic and manual controls, instruments, fittings, safety devices, and accessories needed for safe and efficient operation of the generating and charging plant shall be provided.

3.7.6.1 Instruments. All instruments necessary for indicating the information needed in controlling the equipment and determining its performance shall be grouped into control or instrument panels and securely mounted thereon in positions readily visible and accessible to operation and maintenance personnel.

3.7.6.2 Controls. Insofar as practicable, all manually operated controls used in normal operation of the generating and charging plant shall be grouped into control panels and located in positions readily visible and accessible to an operator standing at the control panel.

3.7.6.2.1 Electrical controls. Heavy-duty, magnetically operated contactors, overload protective devices, and undervoltage protection devices of applicable power-handling capacity shall be provided for each electric motor. Heavy-duty switches or controls and overload protection device shall be provided for all other electrically operated components.

3.7.6.2.1.1 Electric control center. Insofar as practicable, electrical controls, switches, and protective devices, including motor starters, transformers, and circuit breakers shall be grouped into an electric control center, installed at a location convenient to the operators and maintenance personnel. The electric control center shall include a master shutoff and disconnect switch to control all generating and charging plant electric power, and which will shut down all electrical equipment except building lights when open. The electric control center shall also include provisions for wiring in of the building's main power supply and distribution of this power as necessary to all components of the generating and charging plant.

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3.7.6.2.1.2 Position indicators. Manually operated controls that must be closely regulated for generating plant operation shall be equipped with position indicators. The position indicators shall have graduations to permit accurate adjustment of controls to predetermined effective settings.

3.7.7 Parts fabrication. Sheet-metal parts shall be of sufficient thickness, rigidity, and strength to withstand dents, warping, vibration, radiated heat, et cetera, encountered in service, maintenance and under the conditions specified herein.

3.7.8 Locking devices. Where practicable, all screws, pins, bolts, et cetera, shall be equipped with locking devices. Safety wire, self-locking nuts, cotter pins, lockwashers, et cetera, will be acceptable. Where practicable, lockwashers shall be secured to bolts or screws. Cotter pins shall be fabricated from corrosion-resistant steel or other corrosion-resistant material.

3.7.8.1 Pipe thread lubricant and sealant. Tetrafluoroethylene tape conforming to MIL-T-27730, or other pipe thread sealing material specifically approved by the procuring activity, shall be applied to threads prior to assembly of all threaded fittings subject to contact by oxygen of 30 percent or greater purity.

3.7.9 Lifting and tiedown provisions.

3.7.9.1 Lifting rings. Each shipping section shall be provided with lifting rings in locations that will permit cable attachment and lifting without damage to any component. The lifting rings shall be constructed and installed to permit lifting the shipping section when ready for shipment, considering vertical accelerations of 3 gravity units (g). The lifting rings shall be so located on the shipping section that transportation personnel can easily rig safe lifting slings from common cable and spreader bar components for lifting by a single-hook, overhead crane.

3.7.9.2 Lifting eyes. Lifting eyes or other hoisting attachments shall be provided on major components so that they may be readily removed or replaced with overhead hoisting facilities. Insofar as is practicable, all individual components shall be installed in a manner that will permit their removal without necessity for removing other components.

3.7.10 Pressure relief devices. Pressure relief devices shall be provided in any line or component that can be so isolated by closing valves, or other wise, that dangerous pressures could develop. Relief valves shall be provided for normal relief of pressure. All pressure relief devices shall be installed in readily accessible locations. Provisions shall be incorporated to insure discharge of pressure without damage to equipment or danger to personnel. Safety valves shall be sized in accordance with the ASME Unfired Pressure Vessel Code and selected from the list published in Safety Valves and Relief Valves approved by the National Board.

3.7.10.1 Blowout-type safety heads. Blowout-type safety heads may be provided where pressure will not normally have to be relieved, but where protection is necessary, such as for the air processing and separation equipment insulated casings.

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3.7.11 Air processing and separation equipment. The air processing and separation equipment shall be designed and constructed to receive compressed air from the air compression equipment, and to separate therefrom oxygen product and nitrogen product as necessary for compliance with 3.9.1 through 3.9.1.1.

3.7.11.1 The air processing and separation equipment shall consist of the heat exchangers, refrigeration systems, air purification systems, expansion engines (if applicable), air rectification columns, product delivery provisions, hydrocarbon and other absorbent traps, controls, instruments, and accessories necessary for compliance with the requirements specified herein.

3.7.11.1.1 Mechanical refrigeration equipment. Mechanical refrigeration equipment furnished as part of the generating and charging plant shall be designed, constructed, and installed in accordance with the ASRE Safety Code for Mechanical Refrigeration. Any refrigeration system used shall include its own drive, compressors, coolers, condenser, oil traps, evaporator, liquid refrigerant receiver, refrigerant driers, and all controls, instruments, and accessories needed for safe and efficient operation.

3.7.11.1.1.1 Liquid refrigerant receiver. The liquid refrigerant receiver for any mechanical refrigeration system furnished shall be capable of storing the entire refrigerant system refrigerant charge when the generating and charging plant components are stabilized at 160° Fahrenheit (F).

3.7.11.1.1.2 Oil return provisions. Any mechanical refrigeration system with an oil lubricated compressor shall be equipped with oil traps or other provisions as necessary to remove the maximum practicable amount of oil from the refrigerant before it passes to the evaporator and other components operating at low temperature. Means shall be provided for returning oil from the oil traps and from the evaporator to the compressor oil reservoir.

3.7.11.1.1.3 Driers. A desiccant-type drier capable of removing moisture from the liquid refrigerant before it passes to the refrigeration system low temperature components shall be included as part of any mechanical refrigeration system provided.

3.7.11.2 Thawing provisions. The air processing and separation equipment shall include all provisions necessary for thawing if deposits of ice, solid carbon dioxide, or other substances in the processes and effluent passages which cause excessive pressure drops during generating and charging plant operation. Thawing shall require no power or facilities external to the generating and charging plant installation other than the electric service specfied herein for Type I plant.

3.7.11.2.1 Carbon dioxide removal. A standard commercial proven process other, than caustic scrubbing, will be used for carbon dioxide removal.

3.7.11.2.1.2 Crude oxygen purifiers. The generating and charging plant shall have two (2) crude oxygen purifiers to filter carbon dioxide and remove hydrocarbons from the crude liquid oxygen before it enters the low pressure distillation column. Each purifier shall have sufficient capacity for not less than 10 hours of full production capacity of the plant, shall be equipped with filters to prevent adsorbent material or dust from adsorbent material from being blown or carried over into the low pressure column and shall have access openings for draining old adsorbent material and for refilling with new adsorbent material. The two purifiers shall be installed in parallel so that

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when one unit is in service the other unit can be reactivated by passing heated, dry, effluent gases through it. The purifiers shall be connected into the effluent gas discharge system so that effluent gases leaving the heat exchangers can be passed through an electric heater, passed through either purifier, and then discharged to the atmosphere.

3.7.11.3 Insulated casings. All air processing and separation equipment components that normally operate at low temperatures shall be mounted and enclosed within insulated casings for protection and reduction of heat transfer to the components.

3.7.11.3.1 Equipment supports. The insulated casings shall be provided with supports designed to thermally insulate the enclosed equipment from the casing suspension fittings and to retain this equipment in its proper position under all conditions of service. Provisions shall be incorporated to insure protection for the equipment surfaces from damage caused by relative movement of components as a result of temperature variations and stresses that occur during operation and transport.

3.7.11.3.2 Size. The casings shall be large enough to permit installation of the insulation for preventing unnecessary heat leakage to the enclosed equipment.

3.7.11.3.3 Assembly. The casings shall be constructed so that when the air processing and separation equipment individual shipping packages are set in place, the cabinet openings provided for pipe and other connections between insulated casings and assemblies will fit together to provide tight joints, and so that the piping and other connections will be properly aligned.

3.7.11.3.3.1 Interconnecting piping between insulated casings shall be insulated to insure against unnecessary heat leak.

3.7.11.3.3.2 Case openings for piping connections between insulated casings shall be large enough to permit installing enough insulation between the piping joints and the casing metal to insure against unnecessary heat leak.

3.7.11.3.3.3 Shipping covers. The casing openings for interconnecting piping shall be provided with tight-fitting shipping covers to protect the piping connections, and to prevent the entrance of dirt and moisture.

3.7.11.3.4 Removable Panels. The casings shall be constructed with removable panels located and of such size as to permit ready access to the enclosed equipment for inspection, maintenance, and repair.

3.7.11.3.4.1 Sealing. Each panel shall be provided with a sealant for preventing flow of air through the panel joints when the generating and charging plant components change temperature. Panel attachment screws or bolts shall be spaced to insure maintaining good contact between the panel, sealant, and casing at all points under any of the conditions specified herein.

3.7.11.3.5 Relief vent. Each casing shall be equipped with a relief vent to prevent excessive pressure from building up inside. The vent shall be capable of relieving the full gas discharge capacity of the enclosed components in case of rupture without exceeding the safe internal pressure for the casing.

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3.7.11.3.6 Insulation. All space within each casing not occupied by equipment or structural members shall be packed with insulation of a type that is not flammable and that does not readily absorb moisture. The insulation shall be of such a texture that it can be readily removed and replaced by maintenance personnel in the field without the use of special equipment.

3.7.11.3.7 Air bleeds. Air bleeds designed for maintaining a slight positive pressure within the insulated casings by using dry air or nitrogen from the process air or waste nitrogen streams shall be provided.

3.7.11.4 Instruments and controls.

3.7.11.4.1 All normally used operating controls and instruments shall be readily accessible to operating and maintenance personnel without removal of cover plates or panels. Insofar as practicable, controls and instruments shall be so designed and installed that all cold portions are within the insulated casings.

3.7.11.4.1.1 Controls and instruments shall be so installed that air leaks through or around the installations will be effectively prevented.

3.7.11.4.1.2 Only materials that offer considerable resistance to heat flow shall be used to connect internal parts with those extending outside the insulated casings.

3.7.11.4.2 Control valves. All control valves needed for control of the air processing and separation equipment shall be provided.

3.7.11.4.2.1 Wherever practicable, the control valves shall be solidly mounted on the air processing and separation equipment control panels in positions readily visible and accessible to the operating and maintenance personnel. Valves that cannot be mounted on the control panels shall be mounted in readily accessible locations as near the control panels as practicable.

3.7.11.4.2.2 Valve position indicators. All control valves that require setting at some definite position between closed and fully open shall be provided with valve position indicators to permit accurate adjustment to predetermined proper settings.

3.7.11.4.3 Safety devices. All safety devices, including automatic warning signals, automatic shutdown devices, etc., necessary for protection of the air processing and separation equipment against damage in operation shall be provided.

3.7.11.4.3.1 Safety valves. Pop-type safety valves shall be installed in all pressure vessels and lines where a dangerous buildup of pressure could occur. Safety valves handling high-purity oxygen or nitrogen shall be so located and installed that discharge therefrom can be readily piped outside the building.

3.7.11.4.3.1.1 Safety valve mounting. Insofar as practicable, the safety valves shall be grouped in one general location on the air processing and separation equipment, and shall be mounted where they are readily accessible for inspection and maintenance and protected against mechanical damage. The safety valve outlet openings shall be so located and directed that the safety valves will not discharge toward areas normally occupied by operating or maintenance personnel or personnel standing at any of the generating and charging plant and control panels.

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3.7.11.4.4 Instrumentation. All instruments necessary for indicating the temperatures, pressures, liquid levels, and rates used in controlling the generating and charging plant and determining its performance shall be provided. Instruments shall be of the back-connected, flush-mounted type. They shall be securely mounted on the generating and charging plant control panels in positions readily visible and accessible to the operating and maintenance personnel.

3.7.11.4.4.1 Pressure gages shall be provided to indicate all pressures necessary for safe and effective operation, servicing, and maintenance of the generating and charging plant.

3.7.11.4.4.1.1 Pressure gages used for indicating critical pressure such as the air pressure from the compressor, air pressure to and from the expansion engine (if used) and expansion valves, air pressure to and from the heat exchangers, column pressures, etc., shall conform to MIL-G-18997, shall have dial diameters of not less than 4 inches, and shall be accurate to within 1 percent of full-scale range.

3.7.11.4.4.1.2 Pressure gages used for indicating noncritical pressures shall conform to MIL-G-18997, shall have diameters of not less than 2-1/2 inches, and shall be accurate to within 2 percent of full-scale range.

3.7.11.4.4.1.3 Pressure gage range. The indicating range of each pressure gage shall be approximately 1-1/2 times the normal working pressure unless maximum pressures necessitate a higher range to prevent the indicator moving off scale. Each gage shall be graduated in pounds per square inch (psi) over approximately 270° of the dial face. Gages subject to pressures in excess of 100 psi shall have safety backs.

3.7.11.4.4.1.4 A pulsation snubber shall be provided for each pressure gage subjected to pressure pulsations that would otherwise make the gage difficult to read, or that would possibly cause premature gage failure.

3.7.11.4.4.1.5 Shutoff valves. Each of the air processing and separation equipment pressure gages shall be equipped with a shutoff valve that may be throttled to dampen out excessive pulsations of the pressure and that may be completely closed to seal off the line if the gage must be removed during generating and charging plant operation.

3.7.11.4.4.2 Low temperature liquid level gages. Each gage used for indicating the level of a low temperature liquid shall be a differential-pressure-type gage with a dial diameter of not less than 6 inches. Each gage shall be equipped with a zero-setting correction device.

3.7.11.4.4.3 Industrial type dial thermometers with dial diameters of not less than 3 inches, thermocouples, or resistance bulbs, as applicable, shall be provided wherever needed for determining the temperatures used in operating the generating and charging plant air processing and separation equipment.

3.7.11.5 Liquid product delivery provisions. The generating and charging plant air processing and separation equipment shall include separate liquid product delivery provisions for each of the two liquid products.

3.7.11.5.1 The liquid product delivery provisions for each product shall consist of the piping flow control and shutoff valves, pumping equipment provisions and connections, and liquid product delivery connection necessary

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for compliance with the requirements specified herein. Insofar as practicable, all components shall be so designed and installed that all cold portions are within the insulated air separation equipment cold-box casings.

3.7.11.5.2 Liquid product delivery connections. Separate insulated liquid-oxygen and liquid-nitrogen product delivery connections shall be provided on the air processing and separation equipment in protected positions readily accessible for installation of the liquid product delivery lines. The liquid product delivery connections shall be equipped with fittings that will provide for ready and leak tight attachment and ready removal of the liquid product delivery lines, and which are so different that direct connection of the oxygen delivery lines, to the nitrogen delivery connection or of the nitrogen delivery line to the oxygen delivery connection will be impossible.

3.7.11.5.2.1 Each liquid product delivery connection shall be mounted on an insulated plate that will prevent strain on the delivery piping within the air processing and separation equipment during attachment or removal of the liquid product delivery line or while the line is attached. The insulating plate shall be of such size and construction as to insure the minimum practicable heat transfer from structural members, the cabinet casing, or other warm parts to the connection.

3.7.11.5.2.2 Protective cap. Each liquid product delivery connection shall be provided with a protective cap or plug that will tightly close and protect the connection when the liquid product delivery line is not installed. Each protective cap or plug shall be attached to its fitting by a chain or other device to protect against accidental loss.

3.7.11.5.2.3 Oxygen delivery adapter. The adapter shall consist of a 1-inch, male, liquid-oxygen coupling half assembly made up from a coupling seat, gasket and a cap conforming to Drawings C8987855, B8987832, and C8987869, respectively, assembled as shown on Drawing C8987830.

3.7.11.5.2.4 Nitrogen delivery adapter. The adapter shall consist of a 1-inch, male, liquid-nitrogen coupling half assembly made up from a coupling seat, gasket, and a cap conforming to Drawing C8987876, B8987832, and C8987890, respectively, assembled as shown on Drawing C8987888.

3.7.11.6 Gas charging equipment. The generating and charging plant shall include the apparatus, attachments, and accessories necessary for charging oxygen and nitrogen gas into storage containers at any pressure up to and including 3,500 psi as specified in 3.9.1 through 3.9.1.1. Separate and distinct gas charging equipment shall be provided for oxygen gas and for nitrogen gas to insure against mixing or mistaking the product being charged.

3.7.11.6.1 The gas charging equipment for each product shall operate by passing the liquid product from the air separation equipment through a pump inlet filter, pumping the liquid to the necessary pressure, passing the high-pressure liquid through a pump discharge filter and the product vaporizing and warming passages of the air processing and separation equipment heat exchangers and charging the high-pressure gas into storage containers through the gaseous product servicing systems specified.

3.7.11.6.2 The gas charging equipment for each product shall consist of a liquid pump inlet filter, a high-pressure liquid pump, a liquid pump discharge filter, product vaporizing and warming passages in the air processing and separation equipment heat exchangers, a gaseous product delivery connection,

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and the piping valves, fittings, connections, and accessories necessary for operation in accordance with the requirements specified herein.

3.7.11.6.3 Gaseous product delivery connections. Separate gaseous oxygen and gaseous nitrogen product delivery connections shall be provided on the air processing and separation equipment in protected locations readily accessible for installation of the gaseous product delivery lines. The gaseous product delivery connections shall be equipped with fittings that will provide for ready and leaktight attachment and ready removal of the gaseous product delivery lines, and which are so different that direct connection of the oxygen delivery line to the nitrogen delivery connection or of the nitrogen delivery line to the oxygen delivery connection will be impossible.

3.7.11.6.3.1 Each gaseous product delivery connection shall be mounted to prevent strain on the gas charging equipment piping during attachment or removal of the gaseous product delivery lines, or while the line is attached.

3.7.11.6.3.2 Each gaseous product delivery connection shall be provided with a protective cap or plug that will tightly close and protect the connection when the gaseous delivery line is not installed. The protective cap or plug shall be attached to the connection fitting by a chain or other device to protect against accidental loss.

3.7.11.6.4 Controls, instruments, and safety devices. All controls, instruments, safety devices, and accessories necessary for proper and safe operation of the gas charging equipment shall be provided.

3.7.11.7 Charging manifold assemblies. Each charging manifold assembly shall be designed and constructed to charge into high-pressure gas storage containers the oxygen or nitrogen vapors received from the air processing and separation equipment through its gaseous product delivery line.

3.7.11.7.1 Each charging manifold assembly shall consist of a charging manifold, plug the mountings, controls, instruments, and accessories needed for safe and efficient operation of all parts.

3.7.11.7.2 Charging manifolds. Each charging manifold shall consist of two independent charging sections, control valves, and cylinder evacuation provisions. The independent charging manifold sections and the control valves shall be designed and constructed for mounting within the enclosing building on an outside wall in a location readily accessible for carrying empty high-pressure gas storage cylinders to the manifold, charging the cylinders, and removing the charged cylinders from the building.

3.7.11.7.2.1 Manifold sections. Each manifold section shall have seven charging stations installed at approximately the height of the valve connections on standard fifty-one inch oxygen or nitrogen gas storage cylinders and at intervals that will provide space for connecting, disconnecting, and replacing such cylinders standing side-by-side along the manifold section. Each manifold section shall be capable of charging at the full capacity specified in 3.9.1 and 3.9.1.1 when one or all of the charging station valves is open to a receiving container. An automatic check valve shall be provided for each manifold section to prevent backflow from that section.

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3.7.11.7.2.2 Charging station valves. Each individual charging station shall be equipped with a manually operated shutoff valve designed and constructed for handling 3,500 psi pure oxygen gas or nitrogen gas. Each charging station valve shall be capable of passing the full generating and charging plant charging capacity without excessive pressure drop, and shall positively isolate the attaching gas storage container from the manifold section when closed. The valves shall be readily operable by hand, and shall have outlet connections designed for attachment and interchange of the charging leads specified in 3.7.11.8.

3.7.11.7.2.2.1 Each individual charge station shutoff valve shall be so mounted that excessive force exerted in opening or closing the valve and in attaching or removing charging leads or charging hoses will not be applied in the charging manifold piping or fittings.

3.7.11.7.2.2.2 Each charging station valve outlet shall be equipped with a protective cap or plug that tightly closes and protects the fitting when not in use. Each protective cap or plug shall be attached to its disconnect fitting by a chain or other device that will protect against accidental loss.

3.7.11.7.2.2.3 Cylinder Evacuation. Each oxygen manifold section shall have provisions for evacuating seven standard fifty-one inch oxygen cylinder to an absolute pressure of less than three inches of mercury in less than ten minutes and maintaining this pressure until filling begins. One vacuum pump capable of achieving the required vacuum shall be furnished with each generating and charging plant.

3.7.11.8 Charging leads. Fourteen flexible charging leads designed for connecting oxygen cylinder valves with CGA outlet connection number 541 to the oxygen charging station shutoff valves specified in 3.7.11.7.2.2 and 14 flexible charging leads designed for connecting oil-free nitrogen cylinder valves with CGA outlet connection number 581 to the N_2 charging station shutoff valves shall be provided (see Pamphlet V-1).

3.7.11.8.1 The charging leads shall be constructed for handling pure oxygen or nitrogen vapors at pressures up to and including 4,000 psi. The charging leads shall be sufficiently flexible to permit ready connection of the cylinder valves to the charging station valve outlets fittings and to prevent damage from continued flexing in use.

3.7.11.8.2 Pressure-handling capabilities.

3.7.11.8.2.1 Hydrostatic pressure. The charging leads shall not leak or show evidence of damage when subjected to an internal water pressure of 8,000 psi for 5 to 10 minutes.

3.7.11.8.2.2 Gas pressure. The charging leads shall not leak or show evidence of damage when subjected to an internal gas pressure of 4,000 psi for 5 to 10 minutes following the hydrostatic test specified in 3.7.11.8.2.1.

3.7.11.8.2.3 Burst pressure. The charging leads shall not burst when subjected to an internal hydraulic pressure of 16,000 psi.

3.7.11.8.3 The charging leads shall be so designed and constructed that they can be attached or removed with one open-end or adjustable wrench.

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3.7.12 Liquid pump. The high-pressure liquid pump shall be a positive-displacement-type unit capable of handling liquid oxygen or liquid nitrogen and complying with the performance requirements specified herein.

3.7.12.1 The liquid pump shall be designed, constructed, and installed with thermal barriers that will permit the drive mechanism and bearings to operate at temperatures suitable for proper lubrication and long life when the pump head is continually at liquid oxygen or liquid nitrogen temperature. The design, construction, and installation of the liquid pump shall also provide for:

a. Automatic lubrication of all parts requiring same by a self-contained lubrication system. Lubrication reservoirs or sumps shall be of sufficient capacity to insure a sufficient supply of lubricant under all conditions of operation without necessity for frequent additions.

b. Ready inspection, servicing, adjustment, disassembly, repair, and replacement of components under field conditions.

c. Ready detection and correction of packing or seal leakage.

d. Operating packings, seals, shafts, and other moving parts under conditions that will insure against condensation and freezing of water where it will damage packings or seals or interfere with liquid pump operation.

e. Ready adjustment of pump packing while it is in operation (if a packed-typed pump is used).

f. Readily changing the pumping capacity by varying the pump speed.

g. Change from one product to the other without the necessity for replacing the pump packing.

3.7.12.1.1 Fabrication, assembly, and installation of the liquid pump and each of its parts shall be such as to insure operation at full rated capacity without damage, leakage, breakage, cracking, loosening, distortion, or deterioration due to physical stresses.

3.7.12.1.2 The liquid pump shall be contructed of materials not attacked or adversely affected by contact with liquid oxygen, gaseous oxygen, liquid nitrogen, gaseous nitrogen, water, moist air, or cleaning solvents such as trichloroethylene or carbon tetrachloride.

3.7.12.1.3 Liquid pump discharge filters. The liquid pump installation shall include a filter through which all product dispensed from the liquid pump must pass. Each filter shall remove 98 percent, by weight, at operating conditions of all particles whose two smallest dimensions are 10 microns or greater (10 micron nominal rating). Each filter shall further remove all particles whose two smallest dimensions are 40 microns or greater (40 micron absolute rating).

3.7.13 Accessories. All accessories necessary for safe, dependable, and efficient operation of the generating and charging plant shall be provided.

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3.7.13.1 Product quality analysis equipment. Analytical equipment necessary for determining product purity for proper operation and control of the plant shall be provided. The purity analysis equipment shall include the following:

a. Orsat-type oxygen and nitrogen purity analysis kit.

b. Acetylene content test kit.

3.7.13.2 Operational and maintenance tools. A complete set of all special tools, other than common hand tools and those tools normally available in base maintenance shops, shall be furnished for proper operation and maintenance of the generating plant.

3.8 Transportability. Each individual shipping section shall be capable of withstanding forces as imposed by the flight and taxing loads specified in MIL-A-8421 and withstand loads and forces imposed in shipment over highway, railroad or water.

3.9 Performance.

3.9.1 Production capacity. When operating at sea level atmospheric pressure (760 millimeter of mercury), in an ambient temperature (outside the enclosing building) of 70°F, with the intake cooling and process air relative humidity at 50 percent, the generating and charging plant shall be capable of continuously delivering liquid product through the servicing lines and gaseous product through the charging manifold assemblies at a daily average of not less than the rates specified in Table I. Liquid delivery rates shall be based on the quantity of product accumulated in a container maintained at atmospheric pressure.

TABLE I. Delivery capabilities

	Oxygen Liquid	Oxygen Gas	Nitrogen Liquid	Nitrogen Gas
Production capacity	125 lb/hr	1,500 SCFH[1]	84 lb/hr	1,150 SCFH[1]
Delivery pressure	0-5 lb per sq in gage	0-3,500 psi	0-5 lb per sq in gage	0-3,500 psi

[1]SCFH - Standard (760 millimeters of mercury, 70°F) cubic feet per hour

3.9.1.1 Concurrent product delivery. The generating and charging plant shall be capable of concurrently delivering liquid product from the liquid servicing lines and gaseous product from the charging manifold assemblies as follows:

a. Liquid oxygen at 5 pounds per square inch gage (psig) - liquid nitrogen at 5 psig.

b. Liquid oxygen at 5 psig - gaseous nitrogen at 3500 psi.

c. Liquid nitrogen at 5 psig - gaseous oxygen at 3500 psi.

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3.9.1.1.1 During concurrent production, the generating and charging plant shall be capable of delivering each product at a rate reduced from the capacities specified in 3.9.1 by not more than the percentage of full capacity at which the other is delivered.

3.9.1.2 Product quality.

3.9.1.2.1 Liquid oxygen. Liquid oxygen product shall conform to the Type II requirements specified in MIL-O-27210.

3.9.1.2.2 Gaseous oxygen. Gaseous oxygen product shall conform to the Type I requirements specified in MIL-O-27210.

3.9.1.2.3 Liquid nitrogen. Liquid nitrogen product shall conform to the Type II, Class 1, Grade B requirements specified in BB-N-411.

3.9.1.2.4 Gaseous nitrogen. Gaseous nitrogen product shall conform to the Type I, Class 1, Grade B requirements specified in BB-N-411.

3.9.2 Operation.

3.9.2.1 Starting. When all generating and charging plant components are initially at the temperature specified in 3.9.1, the equipment shall be capable of cooling down and achieving full rated production within 5 hours after starting the air compressor.

3.9.2.2 Endurance. The generating and charging plant shall be capable of not less than 30 days continuous production without necessity for plant thaw. Thaw of the heat exchangers will be permitted after a 10-day period of operation if such a thaw does not reduce average production below the rates specified in 3.9.1 and 3.9.1.1. During a heat exchanger thaw, liquid shall be maintained in the air rectification column. Reactivation of carbon dioxide filters and similar periodic normal operational procedures shall not be affected by this requirement. During a 30-day continuous production period, a shutdown of one hour total out of each 24 hours (1 hour each calendar day) will be permitted for lubrication, adjustments, minor repairs, inspections, and preventive maintenance.

3.9.2.2.1 Restart time. Upon resuming operation after the permitted shutdown period, the generating and charging plant shall be capable of resuming rated production within an additional 60 minutes (min) or an additional time equal to the length of the shutdown, whichever is less.

3.9.2.3 Thawing. Upon completion of a 30-day production period, the generating and charging plant shall be capable of being completely thawed, serviced, and inspected, ready for another 30-day production period, within 12 hours.

3.9.3 Environmental conditions. The generating and charging plant shall be capable of operating under and withstanding the following environmental conditions without damage.

3.9.3.1 Operating temperature. The generating and charging plant shall be capable of starting and operating continuously at normal (not necessarily rated) production when the ambient temperature (outside the enclosing building) is between -65° and +125°F, including exposure to solar radiation of 100-120 watts per square foot at the higher temperature.

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3.9.3.2 Humidity. The generating and charging plant shall not be damaged by storage or operating in relative humidities up to and including 100 percent, including conditions wherein condensation takes place in the form of water and frost.

3.9.3.3 Fungus. The generating and charging plant shall not be damaged by exposure to fungus growth such as encountered in tropical and subtropical climates.

3.9.3.4 Salt atmosphere. The generating and charging plant shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard.

3.9.3.5 Rain. The generating and charging plant components installed external to the building shall not be damaged by operation or storage under heavy rainfall as encountered in any locale.

3.9.3.6 Sand and dust. The generating and charging plant shall not be damaged by operation or storage in an atmosphere containing air-borne sand and dust particles such as encountered in desert locations.

3.9.3.7 Wind. The generating and charging plant components installed external to the building shall not be damaged by wind velocities of up to and including 125 miles per hour (mph).

3.9.4 Air compression equipment. The air compression equipment shall function to supply the compressed air needed in the generating and charging plant separation cycle. The air compression equipment shall be a complete, self-contained assembly requiring only a prime mover, lubricants, and cooling-system filling fluid for operation.

3.9.4.1 The air compression equipment shall consist of a single air compression unit including an electric or diesel prime mover, a heavy duty air compressor, intercoolers, aftercooler, radiators, moisture separators, controls, instruments, piping, and all other auxiliary equipment necessary for proper operation. The air compression unit shall be so designed, constructed, and connected to the other equipment, that when operating it will be a complete, compact, and efficient assembly capable of continuously supplying the generating and charging plant air processing and separation equipment with compressed air of the proper purity, volume, temperature, and pressure for compliance with the production requirements specified in 3.9.1 through 3.9.1.1.

3.9.4.2 Mounting. The complete air compression unit, including its prime mover, compressor, intercoolers, aftercooler, controls, instruments, piping, and auxiliary equipment shall be mounted on one or more shipping skids ready for installation on the concrete floor specified in 3.5.

3.9.4.3 Alignment. Provisions shall be made for proper alignment of the air compression equipment components on each skid after installation.

3.9.5 Power transmission system. A power transmission system capable of efficiently transmitting power from the prime mover to the air compressor shall be provided. If V-belt drive is used, provisions for adjusting the belt tension

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by changing the distance between centers of the driving and driven shafts shall be incorporated. Jackscrews ahall be provided as necessary in the base and slide-rail assemblies to facilitate this adjustment.

3.9.6 Air compressor. The air compressor shall be a single, heavy-duty, commercial-type machine of a design and construction that has previously demonstrated a continuous full-load operational capability of not less than 10,000 hours uninterrupted duty without necessity for shutdown. The air compressor shall be equipped with all manual and automatic controls, instruments, filters, lubricating systems, cooling systems, and other accessories needed to insure safe and effective operation during extended periods of continuous usage. The maximum allowable piston speed shall be 700 feet per minute (fpm) for reciprocating compressors.

3.9.6.1 Lubricating system. The air compressor shall be equipped with a self-contained, positive pressure lubricating system capable of thoroughly lubricating all rubber parts requiring lubrication under the conditions specified herein.

3.9.6.1.1 Lubricating oil. The air compressor shall perform as specified herein when supplied with lubricating oil conforming to MIL-L-2104 or MIL-L-26087.

3.9.6.1.2 Upper cylinder lubrication. If required for lubrication of high-pressure cylinders, a standard commercial-type, force-feed lubricator shall be provided. This lubricator shall be driven from a compressor shaft or other positive-acting mechanism.

3.9.6.1.3 Upper cylinder lubricating fluid. The upper cylinder lubricating fluid shall be of the type and viscosity grade specified by the air compressor manufacturer.

3.9.6.2 Cooling system. The air compressor shall be equipped with a complete liquid-type cooling system, including jacketed cylinders and heads, capable of preventing overheating of the air compressor when it is operated continuously at full load and maximum discharge pressure in an ambient temperature of +125°F (outside the building). The cooling system shall function in accordance with the requirements specified herein when filled with antifreeze that will insure the coolant remaining fluid at -65°F. The cooling system shall not use air from inside the building nor discharge air or heat inside the building.

3.9.6.2.1 The cooling system shall include all air-cooled radiators, liquid-cooled oil coolers, coolant pump, automatic control valves (including an automatic coolant-temperature control valve), and other accessories needed to insure safe and effective operation. A manually operated bypass shall be provided to permit manual control in case of failure of the automatic temperature control valve.

3.9.6.3 Filters. The air compressor shall be equipped with intake process air, lubricating oil, and cooling system fluid filters capable of protecting the air compressor components from harmful solid particles.

3.9.6.3.1 Intake process air filter. The intake process air filter shall be a heavy-duty, oil-bath or dry-type unit capable of insuring the removal of harmful particles from the process air before it enters the air compressor. The intake process air filter shall include provisions that will insure its

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effectiveness when used in cold climates where oil will not remain fluid for proper operation, and shall permit ready cleaning or changing of the filter elements when needed.

3.9.6.3.1.1 The intake process air filter shall include all connections, fittings, ducting, and mounting brackets necessary for inducting the process air either from outside the enclosing building at a level not less than 15 feet above the floor, or from within the building, at the discretion of the operating personnel.

3.9.6.3.2 Lubricating oil filters. The lubricating oil filters shall be capable of separating harmful foreign matter from the lubricating oil.

3.9.6.3.3 Cooling system filters. Cooling system filters or strainers shall be provided in the cooling system wherever necessary to prevent solid particles from plugging any component.

3.9.6.4 Instrumentation. The air compressor shall be equipped with all instruments needed for determining whether it is operating properly. The instruments shall be of the back-connected, flush-mounted type. They shall be mounted on the air compressor instrument panel in positions readily visible and accessible to operating and maintenance personnel. The instruments provided shall include the following:

3.9.6.4.1 Pressure gages. Pressure gages having dial diameters of not less than 4 in. shall be provided to indicate the process air discharge pressure for each compression stage in the aftercooler discharge pressure. Each gage shall have an indicating range of approximately 1-1/2 times the normal working pressure. Gages subject to pressures in excess of 100 psi shall have safety backs.

3.9.6.4.1.1 Pressure gages shall be provided to indicate the air compressor lube oil pressure and the coolant pump discharge pressure. Unless maximum pressure necessitates a higher range to prevent the indicator from moving off scale, each gage shall have an indicating range of approximately 1-1/2 times the normal working pressure. Gages subject to pressures in excess of 100 psi shall have safety backs.

3.9.6.4.1.3 Pulsation snubbers. A pulsation snubber shall be provided for each pressure gage subjected to pressure pulsations that would otherwise make the gage difficult to read, or that would possibly cause premature gage failure.

3.9.6.4.2 Temperature indicators. Dial-type temperature indicators shall be provided for indicating the following temperatures:

a. Process air entering each compression stage.

b. Process air leaving each compression stage.

c. Process air leaving the aftercooler.

d. Coolant temperature into compressor from radiator.

e. Coolant temperature from compressor to radiator.

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f. Lube oil temperature into cooler, if applicable.

g. Lube oil temperature from cooler, if applicable.

3.9.6.4.3 Elapsed time indicator. An elapsed time indicator (hourmeter) with a range of not less than 0 to 9,999 hours shall be provided for indicating the air compression equipment operating time.

3.9.7 Alarm system. The air compression equipment shall be provided with an alarm system that will produce visible and audible warnings in case any of the following conditions arise:

a. Low compressor oil pressure.

b. High compressor coolant temperature.

c. High aftercooler discharge temperature.

d. Process air in coolers approaching freezing point of water.

3.9.8 Process air coolers. The air compression unit shall be equipped with a complete, self-contained, air-cooled process-air cooling system consisting of an intercooler between each two compression stages, an aftercooler following the final compression stage, and a moisture separator for each cooler. Automatic shutters or other automatic provisions that will maintain the process air temperature at a level that will prevent freezing of contained water and water vapor in the coolers or moisture separators during low-temperature operation shall be provided.

3.9.8.1 Cooling capacity. Each intercooler and aftercooler shall be capable of insuring a discharge temperature not in excess of 20°F warmer than the ambient temperature, but not less than 35°F, when the generating and charging plant is operated under any of the conditions specified herein.

3.9.8.2 Installation. The process air coolers shall be installed with all air-cooled heat exchange surfaces completely external to the building so that air or heat will not be removed from or discharged into the building.

3.10 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.11 Finishes and protective coatings.

3.11.1 Surface contacting oxygen. All surfaces, parts, fittings, et cetera, of the generating and charging plant that will be in contact with high-purity oxygen shall be thoroughly cleaned with a nonflammable solvent such as trichlorotrifluoroethane or other cleaning agents specifically approved by the procuring activity, to remove all metal shavings, oil, grease, and other foreign material. No other cleaning, priming, or painting with organic materials shall be performed on these surfaces.

3.11.2 Exposed parts and surfaces. All exposed parts and surfaces of the generating and charging plant except parts and surfaces that contact high-purity oxygen and certain corrosion-resistant materials shall be treated as specified herein.

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3.11.2.1 Corrosion-resistant metal parts and surfaces. Metal parts and surfaces of the generating and charging plant that are fabricated from corrosion-resistant materials need not be given protective coatings if such coatings will interfere with their proper functioning.

3.11.2.2 Rubber products. Power transmission drive belts, electric cables, and other similar soft-rubber components shall not be painted.

3.11.3 Chemical treatment and priming of surfaces not subject to high operating temperatures.

3.11.3.1 Ferrous materials. Ferrous parts and surfaces not zinc coated shall be cleaned and chemically treated in accordance with TT-C-490, grade I, grade II, types 1, 4, or 5, or grade III. A primer coat conforming to TT-P-636 or MIL-P-26915 shall then be applied.

3.11.3.1.1 All bolts, screws, nuts, washers, et cetera, shall be cadmium plated in accordance with QQ-P-416, given a phosphate protective coating conforming to DOD-P-16232, or zinc plated in accordance with ASTM-B-633. Cadmium plating shall be type II or III, as applicable.

3.11.3.2 Aluminum materials. Aluminum and aluminum-alloy parts and surfaces shall be cleaned in accordance with MIL-S-5002. A corrosion preventive chemical film conforming to MIL-C-5541, followed by a primer coat conforming to TT-P-1757 shall then be applied. An anodic film conforming to MIL-A-8625 may be used in lieu of the chemical film in accordance with MIL-C-5541.

3.11.3.3 Magnesium materials. Magnesium-base alloy parts and surfaces shall be cleaned and chemically treated in accordance with MIL-M-3171. Two primer coats conforming to TT-P-1757 shall then be applied.

3.11.3.4 Wooden materials. Wooden parts and surfaces shall be treated with sealer conforming to MIL-S-13518.

3.11.3.5 Other surfaces. Other types of surfaces shall be cleaned and chemically treated in accordance with MIL-T-704. A primer coat conforming to TT-P-1757 shall then be applied.

3.11.4 Chemical treatment of surfaces subject to high operating temperatures. Surfaces subject to high temperatures in operation or use shall be cleaned, but not primed, in accordance with 3.11.3.1 through 3.11.3.5.

3.11.5 Finish coats.

3.11.5.1 Parts not subject to high temperatures. All exposed components and surfaces of the generating and charging plant except those parts subject to high operating temperatures shall be finished with two coats of aircraft gray gloss enamel, color 512, conforming to TT-E-489.

3.11.5.2 Parts subject to high temperature. All exposed components and surfaces of the generating and charging plant that are subject to high temperatures during normal operation and use shall be finished with two coats of paint conforming to TT-P-28. Each coating shall be permitted to dry for not less than 24 hours before either applying a second coat or exposure to high temperatures.

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3.12 Marking.

3.12.1 Removable panels. Each removable panel shall be marked on its inner surface with the information necessary to identify its proper location and position.

3.12.2 Product identification. A positive means for quick identification of the product being dispensed through the servicing line disconnect assemblies and through the liquid servicing lines shall be provided in a conspicuous location on each servicing line disconnect assembly and on each liquid servicing line so that no doubt can exist as to which product is being discharged into the receiving container.

3.12.3 Color identification system.

3.12.3.1 Oxygen servicing lines. The oxygen servicing lines shall have 1- inch wide permanently affixed green bands spaced 6 in. apart over a length of not less than 3 feet on each end. Each end shall also be marked with a permanently attached green band on which OXYGEN is permanently imprinted in white block letters not less than 1/2 in. high.

3.12.3.2 Nitrogen servicing lines. The nitrogen servicing lines shall have 1-in. wide permanently affixed red bands spaced 6 in. apart over a length of not less than 3 feet on each end. Each end shall also be marked with a permanently attached red band on which NITROGEN is permanently imprinted with white block letters not less than 1/2 in. high.

3.12.4 Control marking. Unless location makes such identification impractical, all valves, gages, controls, and indicators used in normal operation and control of the generating and charging plant shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, water, light, heat, cleaning fluids, et cetera, will not fade them or cause them to become eradicated or distorted. Securely attached metal tags may be used to identify valves and other controls not normally used for operation and control of the equipment, or which are so located that the application of nameplates would be impractical.

3.12.5 Operating and precautionary instructions. Brief and easily readable operating and precautionary instructions shall be permanently affixed on or near the generating and charging plant air processing and separation equipment control panel in a readily accessible and protected location. The instructions shall be clean, concise, and sufficient to enable operation of the generating and charging plant without damage to the equipment or injury to personnel. Instruction panels shall be made from sheet aluminum or sheet zinc of not less than 0.050 in. thickness, anodized, or etched to produce raised markings with a black or other dark color background, and with a raised border of not less than 1/4 in. width.

3.12.6 Flow diagram. A flow diagram illustrating the generating and charging plant overall processes and equipment shall be permanently affixed on or near the air processing and separation equipment control panel in a readily accessible and protected location. The flow diagram shall be of sufficient size for clear readability, and shall be made from sheet aluminum or sheet zinc as specified in 3.12.5.

3.12.7 Electrical wiring diagram. A complete electrical wiring diagram detailing the generating and charging plant electrical systems shall be

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permanently affixed on or near the air processing and separation equipment control panel in a readily accessible and protected location. The wiring diagram shall be made from sheet aluminum or sheet zinc as specified in 3.12.5.

3.12.8 Transportation data plate. A transportation data plate in accordance with MIL-P-514 shall be securely attached to the outside of each shipping section in a location that will be readily visible and accessible when the shipping package is prepared for shipment. The transportation data plate shall include the following information:

TRANSPORTATION DATA PLATE

Shipping weight	Overall width
Shipping cubage	Overall height
Overall length	C. G. Location

T.O. or T.M. Number

3.12.9 Lifting instruction plate. An instruction plate containing all information necessary for transportation personnel to rig a safe lifting sling from common cable and spreader bar components and to safely lift the shipping section with a 3-g acceleration shall be securely attached to the outside of the shipping section near the transportation data plate and in a location that will be readily visible and accessible when the shipping section is prepared for shipment. The lifting instruction plate shall be of the same material and prepared in the same manner as the transportation data plate.

3.12.10 Lubrication warning plates. Where the use of high-pressure lubrication equipment will cause damage to grease seals or other parts, a warning plate shall be affixed to the equipment in a conspicuous location.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Workmanship. All generating and charging plant parts and accessories shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; cleaning, et cetera.

3.14.1 Cleaning. Components of the generating and charging plant shall be thoroughly cleaned to remove dirt; excess soldering, brazing, and welding flux; welding slag; loose, spattered, or excess solder; metal chips; and other foreign materials before, during, and after assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where

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such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of the generating and charging plant shall be classified as follows:

a. First Article testing (see 4.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Insofar as practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals spaced to insure continued accuracy.

4.3.2 Tools. Special tools and field equipment shall be used to the maximum practicable extent during testing. Utilization of these tools and the field equipment shall be sufficient to determine their usefulness. Instances where the special tools furnished are inadequate shall be recorded in detail.

4.3.3 Instrumentation.

4.3.3.1 Pressures. Pressures and pressure differentials shall be measured by means of the pressure gages and differential pressure meters furnished as part of the generating and charging plant. Barometric pressures shall be measured by means of a properly calibrated mercurial or aneroid barometer.

4.3.3.1.1 Pressures in excess of atmospheric pressure shall be recorded in pounds per square inch gage. Pressures up to atmospheric pressure, including barometric pressure, shall be recorded in either millimeters of mercury of inches of mercury absolute. Differential pressures shall be recorded in inches of water or inches of mercury, as applicable.

4.3.3.2 Temperatures. Temperatures shall be measured by means of appropriately located thermometers, by thermocouples used with calibrated potentiometers, or by resistance bulbs, as applicable. Whenever practicable, the temperature indicating instruments furnished as part of the generating and charging plant itself shall be used.

4.3.3.2.1 Temperatures shall be recorded in degrees Fahrenheit.

4.3.3.3 Rotational speeds. Rotational speeds shall be determined by a directly coupled revolution counter, by a directly coupled tachometer, or by a strobotachmeter. Whenever practicable, the rotational speed indicating instruments furnished as part of the generating and charging plant itself shall be used.

4.3.3.3.1 Rotational speeds shall be recorded in revolutions per minute.

4.3.4 Production rate determination.

4.3.4.1 Liquid product. Liquid oxygen and liquid nitrogen production rates shall be determined by weighing one of the following methods:

(a) Weighing the liquid delivered into a container maintained at atmospheric pressure.

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(b) Using a cryogenic liquid flowmeter accurate to ±0.5 percent certified by, or traceable to, the National Bureau of Standards.

(c) Vaporizing the liquid product into a totaling type gas flowmeter accurate to ±0.5 percent as certified by, or traceable to, the National Bureau of Standards.

In addition to methods (b) and (c) above, a minimum of 12 hours weighing of liquid oxygen production as specified in 4.6.6.3.2 will be mandatory. Throughout a production capacity measurement period, the column liquid levels and generating and charging plant pressures shall be maintained within the limits recommended by the manufacturer.

4.3.4.1.1 Collection container calibration. Containers for collection and weighing of liquid product shall be calibrated for heat leak evaporation losses. The weight of liquid found to evaporate from the container in a length of time equal to the production collection period may be added to the weight collected to determine the true product delivery capacity. Heat leak evaporation loss rate tests shall be made only when the container is not less than half full, and when the filled container has set for not less than 12 hours after filling to permit stabilization of temperatures in the container and its insulation. Allowance shall not be permitted for liquid evaporated in the liquid product servicing systems since they are part of the generating and charging plant.

4.3.4.1.2 Liquid delivery rates shall be recorded in pounds per hour of the liquid actually delivered into the weighing container.

4.3.4.2 Gaseous product. Gaseous oxygen and gaseous nitrogen production rates shall be determined by the increase of pressure in high-pressure gas storage containers being charged to not less than 3,500 psi or by passing the vapors through the charging manifold assemblies specified in 3.7.11.7 at not less than 3,500 psi, expanding the vapors to gas meter inlet pressure after the charging manifold, and measuring the gas volume with a properly calibrated recording or totaling type gas flowmeter designed to handle the product. Throughout a production capacity measurement period, the column liquid levels and generating and charging plant pressures shall be maintained within the limits recommended by the manufacturer.

4.3.4.2.1 Gaseous product delivery rates shall be recorded both as the number of pounds per hour and as the number of standard (760 millimeters of mercury and 70°F) cubic feet per hour charged through the manifold assmbly.

4.3.5 Lubricants.

4.3.5.1 Air compressor.

4.3.5.1.1 Crankcase oil. The air compressor crankcase lubricating oil used for the testing specified herein shall conform to MIL-L-2104 or MIL-L-26087. The viscosity grade shall be as recommended by the air compressor manufacturer.

4.3.5.1.2 Upper cylinder oil. The air compressor upper cylinder lubricating fluid use for the testing specified herein shall be of the type and viscosity grade specified by the air compressor manufacturer.

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4.3.5.2 Miscellaneous lubricants. Other lubricants required for the testing specified herein shall be in accordance with applicable military specification and of the type and grade recommended by the manufacturer.

4.3.5.3 Lubricating oil consumption. Lubricating oil consumption rates shall be determined by measuring the quantity that must be added to maintain a given level in the oil reservoir.

4.3.5.3.1 Lubricating oil consumption shall be recorded as the number of quarts consumed per 24-hr day or the number of quarts consumed per hour, whichever provides more convenient units.

4.3.5.4 Diesel engine. The diesel engine lubricating oil used for testing specified herein shall conform to MIL-L-2104. The viscosity grade shall be as recommended by the engine manufacturer.

4.3.6 Inspections, servicings, and adjustments. Inspections, servicings, and adjustments shall be performed as specified by the generating and charging plant operating and maintenance instruction handbook unless such instructions are contrary to those necessary for complying with the requirements specified herein. Throughout the endurance testing, these inspections, servicings, and adjustments shall be performed during the shutdown periods allowed by 4.6.6.2.

4.3.7 Observations. During the progress of all tests, such characteristics as smoothness of operation, vibrations, ease of control, speed regulation, pressure regulation, freedom from leaks, and the operation of auxiliary equipment shall be observed to determine whether the generating and charging plant complies with all requirements specified herein.

4.3.8 Preparation for test. The generating and charging plant shall be prepared for test by assembly, adjustment, and servicing as specified in the operating and maintenance instruction handbook.

4.3.9 Test data.

4.3.9.1 Correction of data. If the testing specified herein cannot be conducted under the stated conditions, the tests may, upon approval by the procuring activity, be conducted under other conditions and the performance under the specified conditions calculated from the tests results obtained. The manner of calculation shall be demonstrated, and actual test data proving the correctness of the calculation methods used shall be presented for review by the procuring activity.

4.3.9.2 Operational data. Operational data shall be recorded at intervals not to exceed 1 hour. Data recorded shall include the following:

a. Air compression equipment.

(1) Discharge pressure for each stage (psig).

(2) Lubricating oil pressure (psig).

(3) Lubricating oil temperature into cooler (°F).

(4) Lubricating oil temperature from cooler (°F).

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(5) Coolant temperature from radiator (°F).

(6) Coolant temperature into radiator (°F).

(7) Air inlet temperature to each stage (°F).

(8) Discharge air temperature from each stage (°F).

(9) Process air temperature from aftercooler (°F).

(10) Pressure drop across inlet air filter (In H_2O).

(11) Upper cylinder oil feed rate for each cylinder with such lubrication (drops per minute).

b. Air separation equipment.

(1) General.

(a) Process air supply pressure (psig).

(b) Process air supply temperature (°F).

(2) Rectification equipment.

(a) Rectifying column liquid levels (inches).

(b) Rectifying column pressures (psig).

(c) Air pressure to heat exchangers (psig).

(d) Air pressure from heat exchangers (psig).

(e) Air pressure to expansion valves (psig).

(f) Control settings.

(g) Product purities (percent).

(h) All other instrument readings and control settings used in controlling the generating and charging plant and determining its operational characteristics.

(3) Refrigeration equipment.

(a) All pressures, temperatures, flow rates, instrument readings, and control settings used in controlling the generating and charging plant and determining its operational characteristics.

(4) Expansion engines (if used).

(a) Expansion engine speed (rpm).

(b) Lubricating oil pressures.

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(c) Inlet air pressure (psig).

(d) Interstage pressures (if applicable) (psig).

(e) Discharge pressure (psig).

(f) Inlet gas temperature (°F).

(g) Discharge gas temperature (°F).

(h) All other instrument readings and control settings used in controlling the expansion engines and determining their operational characteristics.

(5) Air purification and product purification equipment.

(a) All pressures, temperatures, flow rates, instrument readings, and control settings used in controlling the generating and charging plant and determining its operational characteristics.

(6) Diesel Prime Mover.

(a) Lubricating oil pressure.

(b) Fuel oil pressure.

(c) Coolant temperature.

(d) Crankshaft speed.

4.3.9.3 Production data. Production data (average rate at which liquid or gaseous product was produced during the immediately preceding 2 hours) shall be recorded at the completion of each 2 hours of production.

4.3.9.4 Psychrometric data. Wet- and dry-bulb temperature readings shall be recorded at intervals not to exceed 4 hours during operation.

4.3.9.5 Barometric pressure. The barometric pressure shall be recorded at intervals not to exceed 4 hours during operation.

4.3.9.6 Lubricating fluid consumption. The average lubricating fluid consumption rates for each 24 hour or shorter period of operation shall be recorded at the end of the period.

4.3.9.7 Tolerances.

4.3.9.7.1 Data on weights and liquid flow rates shall be accurate to ±0.5 percent.

4.3.9.7.2 Data on gas and vapor flow rates shall be accurate to within ±1 percent.

4.3.9.7.3 Data on dry-bulb and wet-bulb temperatures for psychrometric determinations shall be accurate to within ±1.0 °F.

4.3.9.7.4 Barometric pressure readings shall be accurate to within ±0.1 millimeter of mercury.

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4.3.9.7.5 Data on rotational speeds shall be accurate to within ±1 percent.

4.3.9.7.6 Data on product purities shall be accurate to within ±0.1 percent.

4.3.9.8 Pressure test record. The pressures at which each generating and charging plant component is tested and the length of time during which it is subjected to the pressures shall be recorded.

4.4 First Article tests.

4.4.1 First Article test sample tested by the contractor. The contractor shall subject one generating and charging plant to the first article testing specified in 4.4.3.

4.4.2 First Article test report. Upon completion of first article testing, the contractor shall prepare a first article test report in accordance with MIL-STD-831.

4.4.3 First Article tests. The first article tests shall consist of all tests described under 4.6 except the individual acceptance test specified in 4.6.7.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each generating and charging plant shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Mechanical inspection (see 4.6.2).

c. Functional check (see 4.6.3).

d. Pressure test (see 4.6.5 thru 4.6.5.1.3).

e. Individual acceptance test (see 4.6.7).

4.6 Test methods.

4.6.1 Examination of product. The overall generating and charging plant and its individual component parts shall be examined for compliance with the requirements specified herein. The examination shall include inspection of materials, workmanship, dimensions, markings, et cetera.

4.6.2 Mechanical inspection. A mechanical inspection of all components and parts of the generating and charging plant for defects and damage shall be conducted. All pertinent data concerning condition, defects of manufacture, damage in transit, and damage through use prior to test shall be recorded.

4.6.3 Functional check. All mechanical parts of the generating and charging plant shall be checked for functioning and shall operate freely and function properly.

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4.6.4 Dimensions and weight. The overall dimensions and total weights of each individual shipping section when prepared for shipment shall be determined and recorded.

4.6.5 Pressure test. Each component and circuit of the generating and charging plant that operates under positive pressure in normal service shall be pressure tested in accordance with ASME Unfired Pressure Vessel Code. Automatic pressure relief valves and blowout-type safety heads need not be installed for this testing. Distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

4.6.5.1 Charging lead and charging hose pressure tests.

4.6.5.1.1 Hydraulic pressure test. Each charging lead specified in 3.7.11.8 shall be subjected to an internal water pressure of 8,000 psi for 5 to 10 minutes. Each item shall be observed and examined for leakage or evidence of damage during and after the test. Leakage or damage detected as a result of the test shall be cause for rejection.

4.6.5.1.2 Gas pressure test. After subjection to the test specified in 4.6.5.1.1, each charging lead and each charging hose shall be subjected to an internal gas pressure of not less than 4,000 psi for 5 to 10 minutes. Each item shall be observed for leakage by submersion in a water bath throughout the test, and examined for damage after the test. Leakage or damage detected as a result of the test shall be cause for rejection.

4.6.5.1.3 Burst pressure test. One charging lead from each lot of 50 or less and one charging hose from each lot of 20 or less shall be selected at random and subjected to a sufficient internal hydraulic pressure to cause failure. If the item fails below 16,000 psi, it shall be considered to be unsatisfactory and five more samples from same lot shall be tested in the same manner. If any of the five additional samples fails below 16,000 psi, the entire lot shall be rejected.

4.6.6 Performance demonstration and endurance test. The generating and charging plant shall be subjected to a combination performance demonstration and endurance test consisting of a continuous production period conducted as follows:

4.6.6.1 Preparation for test. The generating and charging plant shall be erected and installed as directed by the procuring activity and run-in as the manufacturer determines necessary for his preliminary checkout operation. All necessary adjustments other than normal control adjustments shall be made during the run-in and shall remain undisturbed thereafter. Following the running-in, the generating and charging plant shall be thawed and serviced for normal operation as specified in the operation and maintenance instruction handbook.

4.6.6.2 Operating conditions. Operation throughout the performance demonstration and endurance test period shall be as continuous as possible, except that the generating and charging plant may be shut down for no longer than 1 hour out of each 24 hours (1 hour each calendar day) if necessary for lubrication, minor adjustments and repair, inspections, and preventive maintenance without penalty. Thaw of the heat exchanger at 10-day intervals (a maximum of three individual component thaws throughout the performance

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demonstration and endurance test period) will be permitted if such heat exchanger thaw procedure does not reduce average production below the rates specified in 3.9.1 and 3.9.1.1. During a heat exchanger thaw, liquid shall be maintained in the air rectification column. The operation time required to achieve the production rate specified in 3.9.1 at the beginning of the test shall not exceed the limit specified in 4.6.6.3.1. The operation time required to resume the production rates specified in 3.9.1 or 3.9.1.1 after each shutdown shall not exceed 1 hour or a length of time equal to the shutdown period, whichever is less. The length of operation time required to achieve rated production at the beginning of the test, the length of operation time required to resume rated production after each shutdown, the length of each shutdown, the reason for each shutdown, corrective actions taken, and all difficulties and malfunctions encountered shall be individually recorded on the test data sheets.

4.6.6.2.1 If an interruption of greater length than that allowed in 4.6.6.2 (totaling more than 1 hour in a 24-hour period) is necessary and results in a production break of 12 hours or less, the required production time shall be extended by 2 hours for each hour shutdown time plus the operation time necessary for resuming the production rates specified in 3.9.1 and 3.9.1.1. Example: a 5-hour shutdown requiring an additional 3 hours of operation to resume rated production would extend the test period by 13 hours of actual production time in addition to that specified herein. If case of necessity for a shutdown in excess of the permissible 1 hour, this 1 hour shall not be omitted from the shutdown time total when computing the specified extension or required production time. If a production break of more than 12 hours is necessary, the purpose of the test shall be considered defeated and the test shall be repeated unless continuation is authorized by the procuring activity.

4.6.6.2.2 Production periods of less than 12 hours duration between production breaks shall not be credited to performance demonstration and endurance test time.

4.6.6.2.3 Operation throughout the performance demonstration and endurance test shall be under conditions as close to the design criteria specified in 3.9.1 as possible consistent with the installation and erection requirements imposed by the procuring activity.

4.6.6.2.4 Throughout the specified performance demonstration and endurance testing specified herein, the generating and charging plant shall deliver liquid product through the liquid servicing lines into storage containers maintained at atmospheric pressure, as specified in 4.3.4.1. Gaseous product shall be delivered through the charging manifold assemblies specified in 3.7.11.7 into high-pressure gas storage containers or through gas meters as specified in 4.3.4.2.

4.6.6.3 Test procedures.

4.6.6.3.1 Concurrent liquid oxygen and gaseous nitrogen production. When all components within the enclosing building are initially at or above the temperature specified in 3.9.1, the generating and charging plant shall be adjusted for simultaneous liquid oxygen and gaseous nitrogen production, started and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rates at which it will continuously and concurrently produce liquid oxygen of the specified quality and gaseous nitrogen of the specified quality have been established for each of the following ratios:

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Ratio	Percent of Maximum Liquid Oxygen Production	Percent of Maximum Gaseous Nitrogen Production
a.	80 to 90	20 to 10
b.	45 to 55	55 to 45

4.6.6.3.1.1 The length of operation time after starting of the air compressor necessary to bring the generating and charging plant into production at the rate required by 3.9.1.1, the maximum liquid oxygen and gaseous nitrogen production rates achieved for each of the specified ratios, and the average liquid oxygen and gaseous nitrogen production rates for each of the specified ratios shall be individually determined and recorded for the test. The operation time required after starting of the air compressor to bring the generating and charging plant into production at the rate required by 3.9.1.1 shall not exceed 5 hours, and the average production rate (based on the daily averages) for each of the specified ratios shall not be less than that required by 3.9.1.1. Liquid oxygen and gaseous nitrogen production shall be continued at maximum rate for not less than 100 hours at each of the specified ratios after reaching rated production capacity for that ratio. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1.1 shall not be credited toward completion of the required production time.

4.6.6.3.2 Liquid oxygen production. Upon completion of the concurrent liquid oxygen and gaseous nitrogen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce liquid oxygen alone, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rate at which it will continuously produce liquid oxygen of the specified quality has been established. The time required to make the changeover from concurrent liquid oxygen and gaseous nitrogen to liquid oxygen production alone, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1, the maximum production rate achieved, and the average production rate shall be individually determined and recorded for the test. The average production rate (based on the daily averages) shall be not less than 125 pounds per hour (pph). Liquid oxygen production shall be continued at maximum rate for not less than 100 hours after reaching rated production capacity. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1 shall not be credited toward completion of the required production time.

4.6.6.3.3 Gaseous oxygen production. Upon completion of the liquid oxygen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce gaseous oxygen alone, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rate at which it will continuously produce gaseous oxygen of the specified quality has been established. The time required to make the changeover from the concurrent liquid oxygen and gaseous oxygen to gaseous oxygen production alone, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1, the maximum production rate achieved, and the average production rate shall be individually determined and recorded for the test. The average production rate (based on the daily averages) shall be not less than 1,500 standard cubic feet per hour. Gaseous oxygen production shall be continued at maximum rate for not

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less than 72 hours after reaching rated production capacity. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1 shall not be credited toward completion of the required production time.

4.6.6.3.4 Gaseous nitrogen production. Upon completion gaseous oxygen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce gaseous nitrogen alone, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rate at which it will continuously produce gaseous nitrogen of the specified quality has been established. The time required to make the changeover from concurrent gaseous oxygen and gaseous nitrogen production to gaseous production alone, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1, the maximum production rate achieved, and the average production rate shall be individually determined and recorded for the test. The average production rate (based on the daily averages) shall not be less than 1,150 standard cubic feet per hour. Gaseous nitrogen production shall be continued at maximum rate for not less than 72 hours after reaching rated production capacity. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1 shall not be credited toward completion of the required production time.

4.6.6.3.5 Liquid nitrogen production. Upon completion gaseous nitrogen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce liquid nitrogen alone, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rate at which it will continuously produce liquid nitrogen of the specified quality has been established. The time required to make the changeover from concurrent gaseous nitrogen and liquid nitrogen production to liquid nitrogen production alone, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1, the maximum production rate achieved, and the average production rate shall be individually determined and recorded for the test. The average production rate (based on the daily averages) shall be not less than 84 pph. Liquid nitrogen production shall be continued at maximum rate for not less than 100 hrs after reaching rated production capacity. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1 shall not be credited toward completion of the required production time.

4.6.6.3.6 Concurrent liquid nitrogen and gaseous oxygen production. Upon completion of the liquid nitrogen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce liquid nitrogen and gaseous oxygen simultaneously, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rates at which it will continuously and concurrently produce liquid nitrogen of the specified quality and gaseous oxygen of the specified quality have been established for each of the ratios required by 4.6.6.3.1. The time required to make the changeover from the liquid nitrogen to concurrent liquid nitrogen and gaseous oxygen production, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1.1 for each of the specified ratios, the maximum liquid nitrogen and gaseous oxygen production rates achieved for each of the specified ratios, and the average liquid nitrogen and gaseous oxygen production rates for each of the

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specified ratios shall be individually determined and recorded for the test. The average production rate (based on the daily averages) for each of the specified ratios shall be not less than that required by 3.9.1.1. Liquid nitrogen and gaseous oxygen production shall be continued at maximum rate for not less than 72 hours at each of the specified ratios after reaching rated production capacity for that ratio. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1.1 shall not be credited toward completion of the required production time.

4.6.6.3.7 Concurrent liquid oxygen and liquid nitrogen production. Upon completion of the concurrent liquid nitrogen and gaseous oxygen production test and without stopping or discontinuing operation, the generating and charging plant shall be changed over to produce liquid oxygen and liquid nitrogen simultaneously, and operated in accordance with 4.6.6.2 through 4.6.6.2.4 until the rates at which it will continuously and concurrently produce liquid oxygen of the specified quality and liquid nitrogen of the specified quality have been established for each of the ratios required by 4.6.6.3.1. The time required to make the changeover from concurrent liquid nitrogen and gaseous oxygen to concurrent liquid oxygen and liquid nitrogen production, the length of operation time necessary to bring the generating and charging plant into production at the rate required by 3.9.1.1 for each of the specified ratios, the maximum liquid oxygen and liquid nitrogen production rates achieved for each of the specified ratios, and the average liquid oxygen and liquid nitrogen production rate for each of the specified ratios shall be individually determined and recorded for the test. The average production rate (based on the daily averages) for each of the specified ratios shall be not less than that required by 3.9.1.1. Liquid oxygen and liquid nitrogen production shall be continued at maximum rate for not less than 100 hours at each of the specified ratios after reaching rated production capacity for that ratio. Operation time during which product is not being delivered at an average rate equal to or above the production rate required by 3.9.1.1 shall not be credited toward completion of the required production time.

4.6.6.4 Thawing. Immediately upon completion of the performance demonstration and endurance test, the generating and charging plant shall be shut down, completely thawed, and serviced as specified in the operation and maintenance handbook for another approximately 30-day production period. The time required to complete the thawing and servicing shall not exceed that specified in 3.9.2.3 and shall be recorded.

4.6.6.5 Evacuation test. The evacuation capability of the oxygen manifold sections will be demonstrated by connecting seven standard fifty-one inch oxygen cylinders to the manifold while at atmospheric pressure. Failure of the pump to reduce the pressure within the manifold and cylinders to less than three inches of mercury absolute within ten minutes after the pump is started shall be cause for rejection.

4.6.7 Individual acceptance test. The generating and charging plant shall be erected and installed as directed by the procuring activity (see 6.2), adjusted for simultaneous liquid oxygen and gaseous nitrogen production at ratio (a) specified in 4.6.6.3.1, started and operated under as close to the design conditions specified in 3.9.1 as practicable and in accordance with the operating conditions specified in 4.6.6.2 through 4.6.6.2.2 for not less than 100 hours of continuously delivering liquid oxygen and gaseous nitrogen under

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the conditions specified in 4.3.4.1 and 4.3.4.2 at full production capacity. Operation time during which liquid oxygen and gaseous nitrogen are not actually being delivered at or above the production rates specified in 3.9.1.1 shall not be credited toward completion of the 100-hour period. All difficulties and malfunctions encountered shall be recorded in detail on the test data sheets.

4.6.8 Environmental testing. The generating and charging plant shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810.

4.6.8.1 Humidity. Equipment samples representing applicable portions of the generating and charging plant that might be damaged by exposure to humidity shall be subjected to, but not operated in, a humidity test in accordance with the procedure specified.

4.6.8.2 Fungus. Equipment samples representing applicable portions of the generating and charging plant that might be damaged by exposure to fungus attack shall be subjected to a fungus-resistance test in accordance with the procedures specified.

4.6.8.3 Salt spray. Equipment samples representing applicable portions of the generating and charging plant that might be damaged by exposure to a salt atmosphere shall be subjected to a salt spray test in accordance with the procedure specified.

4.6.8.4 Sand and dust. Each individual shipping package or components thereof that might be damaged by sand and dust shall be subjected to, but not operated in, a sand and dust test in accordance with the procedure specified.

4.6.8.5 Rain. Each component that will be installed external to the enclosing building shall be subjected to a rain test in accordance with the procedure specified.

4.6.8.6 Operating temperatures and windloads. Design criteria, analyses, or computations shall be submitted to the procuring activity that will substantiate the capability of the generating and charging plant to withstand and function in accordance with the requirements specified herein under the following conditions:

a. Ambient temperature (outside the enclosing building) of -65°F.

b. Ambient temperature (outside the enclosing building) of 125°F.

c. Windloads as specified in 3.9.3.7 (applicable only to the components installed external to the enclosing building).

4.6.9 Servicing and maintenance tests. All normal preventive maintenance and servicing operations specified in the operating and maintenance instruction handbook shall be performed to determine their adequacy, ease of accomplishment and the accessibility of parts and assemblies for performance of same. Insofar as practicable, these operations shall be conducted as part of the normal preventive maintenance, servicings, and inspections performed in accomplishing the testing specified herein. Interferences and obstructions to servicing or preventive maintenance shall be recorded in detail on the test data sheets.

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4.6.10 Mechanical check. Upon completion of the tests specified in 4.6.5 through 4.6.8, a critical inspection shall be made of components to determine their operability and any damage or undue wear incurred during the tests. Teardown and parts measurement shall be made only in those cases where service life is in question. Where teardown and parts measurement are performed, wear or distortion that exceeds limits permitted by the manufacturer for new parts shall be cause for considering the part of parts affected as having failed to satisfactorily complete the test.

4.6.10.1 Steel parts subject to high stress in operation and which are suspected of having defects shall also be subjected to a magnetic particle inspection (magnaflux) and shall exhibit no indications or damage attributable to the tests, or shall be considered as having failed the test. Magnetic particle inspection shall be performed in accordance with MIL-I-6868.

4.6.10.2 Nonmagnetic parts suspected of defects shall be subjected to inspection with fluorescent penetrant (blacklight) and shall exhibit no defects that would impair their life or usefulness. Penetrant inspection shall be perfromed in accordance with MIL-I-6866.

4.7 Inspection of preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The generating and charging plant, parts, fittings, components and assemblies, that during operation, come in direct contact with high-purity oxygen, liquid or gaseous, shall be thoroughly cleaned and dried, by one, or a combination of the following methods:

Method A. Vapor degreasing process followed by fingerprint removal. Use trichlorotrifluoroethane conforming to MIL-C-81302, in a standard commercial vapor degreaser. The operation of the degreaser shall be in accordance with the manufacturer's recommendations. Fingerprint remover shall conform to MIL-C-15074. Following application of the fingerprint remover, surfaces shall be thoroughly rinsed with clean trichlorotrifluoroethane. All solvents shall be removed after the degreasing operation by purging with hot filtered oil-free air or nitrogen, or by vacuum evacuation.

Method B. Solvent degreasing by slushing methods. All surfaces requiring degreasing shall be thoroughly washed with solvent conforming to MIL-C-81302. Degreased surfaces shall be cleaned with fingerprint remover conforming to MIL-C-15704. Following application of the fingerprint remover, the surfaces shall be thoroughly rinsed with clean trichlorotrifluoroethane. All solvents shall be removed after the degreasing operation by purging with hot filtered oil-free air or nitrogen, or by vacuum evacuation. Precautions shall be taken to insure the chlorinated solvent does not contact plastics or other incompatible materials.

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Method C. Alkali detergent degreasing. Components shall be flushed with hot alkaline cleaning compound conforming to P-C-436, until free from oil, grease, and other combustible materials. Following alkaline cleaning, all surfaces, internal and external, as applicable, shall be thoroughly rinsed with clean hot water and dried by purging with hot, filtered oil-free air or nitrogen or by baking at a temperature of 250° to 300°F, until all water is removed.

Method D. Steam cleaning. Cleaning shall be effected by subjecting the component to a stream of steam only or to a stream of steam with an added cleaning compound followed by steam alone. Cleaning compound shall conform to P-C-437. Following steam cleaning, all surfaces, internal and external, as applicable, shall be purged with hot filtered oil-free air or nitrogen or by baking at a temperature of 250° to 300°F, until all water is removed.

Method E. Detergent cleaning. This procedure is applicable to plastics, O-rings, packing, and similar materials. Components shall be cleaned with 4 percent detergent solution (mix 6 ounces of detergent with one gallon of water) for 30 minutes at 120°F. Detergent shall conform to MIL-D-16791. After cleaning rinse with distilled water several times. Dry by purging with hot filtered oil-free air or nitrogen.

Method F. Ultrasonic cleaning. Many cleaning solutions can be used as an ultrasonic cleaning media. These liquids may be either water base detergents or organic solvents. Care shall be taken to insure that the solution will not attack or erode the parts being cleaned. Water base detergent solutions shall be used for cleaning simple unchanneled parts. Detergent shall conform to MIL-D-16791, Type I. Trichlorotrifluoroethane shall be used for cleaning complex parts. Trichlorotrifluoroethane shall conform to MIL-C-81302.

Method G. Oxygen systems cleaning compound. Components shall be rinsed or flushed with cleaning compound conforming to MIL-C-81302. After cleaning, components shall be dried with filtered oil-free air or nitrogen.

5.1.1.2 Parts that do not contact high-purity oxygen shall be cleaned and dried in accordance with the applicable processes and procedure of MIL-P-116 or as specified herein.

5.1.1.3 Unit Packs. Small components that contact high-purity oxygen shall be preserved Method C of MIL-P-116. Prior to inserting into the barrier bag, sharp surfaces and other irregular surfaces of the part shall be cushioned as required to prevent rupture or other damage to the bag. Folded pads, thick wraps, or twisted lengths used as cushioning shall conform to L-P-378 or equal. The cushioning shall be secured in place with vinyl or nylon cord. Prior to heat sealing, the bags shall be purged with filtered oil-free, dry nitrogen. Large components which contact high purity oxygen shall be purged with filtered oil-free dry nitrogen and all openings sealed with blind flanges, expansion plugs, metal caps, polyethylene caps, or polyethylene film conforming to L-P-378 or equal. Sealing shall be accomplished in a manner to provide a positive seal which will prevent entrance of harmful foreign matter. Large or small components that contact high purity oxygen that cannot be preserved as specified herein, shall be preserved Method II in accordance with MIL-P-116.

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5.1.1.4 Instrument Panels. The glass fronts of instruments, instrument panels and dial glasses shall be protected by covering with neutral barrier material conforming to MIL-P-130 or MIL-P-17667. Glasses shall be bridged with pieces of fitted plywood or other similar protective material secured in place with tape conforming to PPP-T-60 or PPP-T-97.

5.1.1.5 Purity test sets: Purity test set components shall be individually wrapped with material conforming to MIL-P-130 or MIL-P-17667 and cushioned with cushioning material conforming to PPP-C-1752 or PPP-C-1797. Glassware shall be individually packaged in interior containers conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676. Containers shall be closed in accordance with the appendix of the container specification.

5.1.1.6 Electric motors and generators. Motors and generators shall be preserved Method II in accordance with MIL-P-116.

5.1.1.7 Push button remote control stations. Push button remote control stations shall be packaged Method II in accordance with MIL-P-116.

5.1.1.8 Electric cables. Electric cables shall be coiled to the minimum safe dimensions and secured in not less than 3 places with tape conforming to PPP-T-97. Exposed ends of wires and sockets shall be sealed with plugs or caps conforming to MIL-C-52078 or wrapped with barrier material conforming to MIL-P-130 or MIL-P-17667 and secured in place with tape conforming to PPP-T-60, Class 1.

5.1.1.9 Diesel engine. The diesel engine shall be preserved and packaged in accordance with MIL-E-10062, Method IIA.

5.1.1.10 Pulleys. All unpainted pulleys shall be coated with compound conforming to TT-P-664, prior to belt installation. Belts shall be placed in operating position, whenever practicable, and tension completely removed. Removed belts shall be wrapped individually for single belt drives, or in sets for multiple drives with barrier material conforming to MIL-B-121, Grade A, Type I and sealed with a heat sealer or pressure sensitive tape conforming to PPP-T-60. Packaged belts shall be secured to the equipment near pulleys on which they will be used. Waterproof tags, conforming to UU-T-81, shall be attached indicating the belts proper application.

5.1.1.11 Maintenance Tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.12 Air compressors. The air compressors shall be preserved and packaged in accordance with MIL-P-3600.

5.1.1.13 Refrigeration and cooling equipment shall be preserved and packaged in accordance with MIL-R-3593.

5.1.1.14 Valves Valves, except valves that contact high-purity oxygen, shall be preserved and packed in accordance with MIL-V-3.

5.1.1.15 Piping and tubing. The ends of small diameter piping and tubing shall be capped with plastic caps or plugs conforming to MIL-C-52078. Desiccant shall be placed in all piping and tubing that contacts high-purity oxygen whenever the diameter is of sufficient size to admit fractional quantities.

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The ends of large diameter piping and tubing which contact high-purity oxygen shall be sealed with gasketed covered plates, metal covers, or blind flanges. As an alternate, sealing can be accomplished by sealing the ends with barrier material conforming to MIL-B-131, Class 1, secured with tape conforming to PPP-T-60. The taped edges shall be oversprayed with strippable coating in accordance with MIL-C-23760. Steel piping and tubing, that does not contact high purity oxygen, shall be preserved and packaged in accordance with MIL-STD-163. Aluminum piping and tubing that does not contact high-purity oxygen shall be preserved and packaged in accordance with MIL-STD-649.

5.1.2 Level C. Items which contact high-purity oxygen shall be perserved and packaged in the same manner as specified for Level A. Items that do not contact high-purity oxygen shall be preserved and packaged in a manner which affords adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirement of this level.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, components of the generating and charging plant, preserved and packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, overseas type, PPP-B-621 Class 2, MIL-C-104, MIL-C-3774, MIL-C-9897, MIL-C-11264, MIL-C-52950. The gross weight of each container shall not exceed the authorized weights specified in the container specification. Containers shall be closed and strapped in accordance with the applicable containers specification and appendix thereto.

5.2.2 Level B. Unless otherwise specified components of the generating and charging plant, preserved and packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, domestic type, PPP-B-621, Class 1, PPP-B-636, Type SF, Class Weather Resistant PPP-B-640, Class 2, Grade A, MIL-C-104, MIL-C-3774, MIL-C-9897, MIL-B-11264 or MIL-C-52950. The gross weight of each container shall not exceed the authorized weight specified in the container specification Containers shall be closed and strapped in accordance with the applicable containers specification and appendix thereto.

5.2.3 Level C. Components of the generating charging plant perserved and packaged as specified in 5.1 shall be packed in a manner which provides adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Physical protection. Contents of shipping containers shall be anchored, blocked braced, and cushioned in accordance with MIL-STD-1186. Testing: freefall drop test, edge-wise drop test, cornerwise drop test, and guided impact tests shall be in accordance with MIL-STD-1186. The tests are not required when level C packing is specified.

5.4 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition, to facilitate reassembly, parts that are removed and mating parts on the basic assembly, shall be matchmarked with lusterless white enamel, overcoated with varnish conforming

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to TT-V-121. Small parts that cannot be matchmarked as specified above shall be matchmarked with waterproof tags conforming to UU-T-81, Class 2.

6. NOTES

6.1 Intended use. The generating and charging plant covered by this specification is intended for use in producing high-purity oxygen or nitrogen for support of aircraft servicing requirements.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether an erection instruction manual illustrated with photographs and isometric or perspective line drawings showing clearly each successive step in field erection of the generating and charging plant should be furnished with each unit (see 3.5.3.3).

c. Whether instruction manuals covering operation, servicing, and maintenance of the generating and charging plant, or other technical (Publications including quantities and disposition) should be provided with each unit.

d. Applicable levels of preservation-packaging and packing required (see 5.1 and 5.2).

e. That the contractor shall, within the time specified by the contract or purchase order, furnish complete foundation and anchoring details for all components of the generating and charging plant requiring special foundation and anchoring.

f. That the contractor shall, within the time specified by the contractor or purchase order, furnish complete details concerning electric power, lighting, water, and other utilities that will be required during erection, test, normal operation and maintenance of the equipment.

g. Liquid product servicing hoses shall be furnished by the Air Force.

6.3 Definitions. For the purpose of this specification, the following definitions shall apply.

6.3.1 Ambient conditions. Ambient conditions are the dry-bulb temperature, wet-bulb temperature, and the relative humidity of the atmospheric air surrounding, and in the vicinity of the generating and charging plant installation, but unaffected by any heat or cold emanating from the equipment.

6.3.2 Production period. A production period is defined as a period of time during which oxygen and nitrogen products are actually being delivered at full capacity into receiving containers.

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6.3.3 Pressures. Unless specified as absolute or denoted by the symbol psia, all pressures referred to herein will be interpreted as pounds per square inch (psi) gage.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Air Force - 99

Reviewer:

DLA-GS

Preparing activity:

Air Force-82

(Project 3655-F098)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-27253C	GENERATING AND CHARGING PLANT, OXYGEN - NITROGEN, SKID-MOUNTED, A/E26A-8 AND A/E26A-20

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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